Practical 2: State Management and Routing in React

Problem Definition

- Extend the portfolio application from Practical 1 by adding React Router.
- Implement a navigation bar with at least 3 routes: Home, Projects, and Contact.
- Each route must render a distinct component without a full page reload.
- Add at least 2 useState variables used meaningfully — one for toggling UI visibility and one for managing a form input on the Contact page.
- The Contact page must include a controlled input that captures and displays user input in real time.
- Add a 404 Not Found route that renders a custom error component for unknown paths.
- Implement a dark/light mode toggle using useState that applies a CSS class to the root element.
- Store the contact form input in state and display a live character count below the input.

Implementation and Actual Code Snippets

File: src/main.jsx

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import { BrowserRouter } from 'react-router-dom';
import App from './App';
import './index.css';

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode>
    <BrowserRouter>
      <App />
    </BrowserRouter>
  </React.StrictMode>
);
```

File: src/App.jsx

```
import React, { useState } from 'react';
import { Routes, Route, NavLink } from 'react-router-dom';
import Home from './pages/Home';
import Projects from './pages/Projects';
import Contact from './pages/Contact';
import NotFound from './pages/NotFound';

export default function App() {
```

```
  const [theme, setTheme] = useState('dark');

  const toggleTheme = () => {
    setTheme((prev) => (prev === 'dark' ? 'light' : 'dark'));
  };

  return (
    <div className={`app-wrapper ${theme}-mode`}>
      <nav className="navbar">
        <div className="nav-brand">AK Nursumar</div>
        <div className="nav-links">
          <NavLink to="/" end className={({ isActive }) => (isActive ? 'active' :
'')}>Home</NavLink>
          <NavLink to="/projects" className={({ isActive }) => (isActive ? 'active' :
'')}>Projects</NavLink>
          <NavLink to="/contact" className={({ isActive }) => (isActive ? 'active' :
'')}>Contact</NavLink>
          <button className="theme-toggle" onClick={toggleTheme}>
            {theme === 'dark' ? '☀ Light' : '🌜 Dark'}
          </button>
        </div>
      </nav>
      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/projects" element={<Projects />} />
        <Route path="/contact" element={<Contact />} />
        <Route path="*" element={<NotFound />} />
      </Routes>
    </div>
  );
}
```

File: src/pages/Contact.jsx

```
import React, { useState } from 'react';

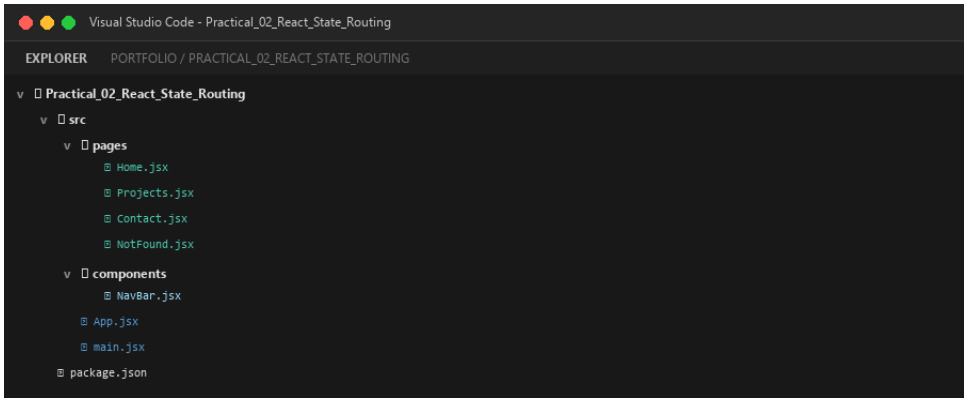
export default function Contact() {
  const [message, setMessage] = useState('');
  const [showTooltip, setShowTooltip] = useState(false);
  const MAX_CHARS = 300;

  return (
    <div className="contact-page card">
      <h2>Contact AK Nursumar</h2>
      <div className="form-group">
        <label>
          Your Message:
          <button type="button" onClick={() => setShowTooltip(!showTooltip)}
className="btn-info">
            ℹ Tooltip
          </button>
        </label>
        {showTooltip && <p className="tooltip">Controlled input updates React state on
```

```
every keystroke.</p>}
        <textarea
          rows="4"
          value={message}
          maxLength={MAX_CHARS}
          onChange={(e) => setMessage(e.target.value)}
          placeholder="Type your message here..."
        />
        <div className="char-count">{message.length} / {MAX_CHARS} characters</div>
      </div>
      <div className="live-preview">
        <h4>Real-Time Controlled State Echo:</h4>
        <p>{message || '(No input yet... start typing above)'}</p>
      </div>
    </div>
  );
}
```

Output

Figure 1: Project Folder Structure



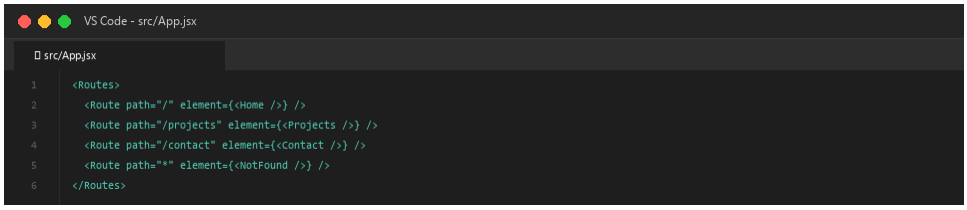
Explanation: *Updated project structure featuring a dedicated src/pages directory housing Home, Projects, Contact, and NotFound route components.*

Figure 2: main.jsx – BrowserRouter Wrap



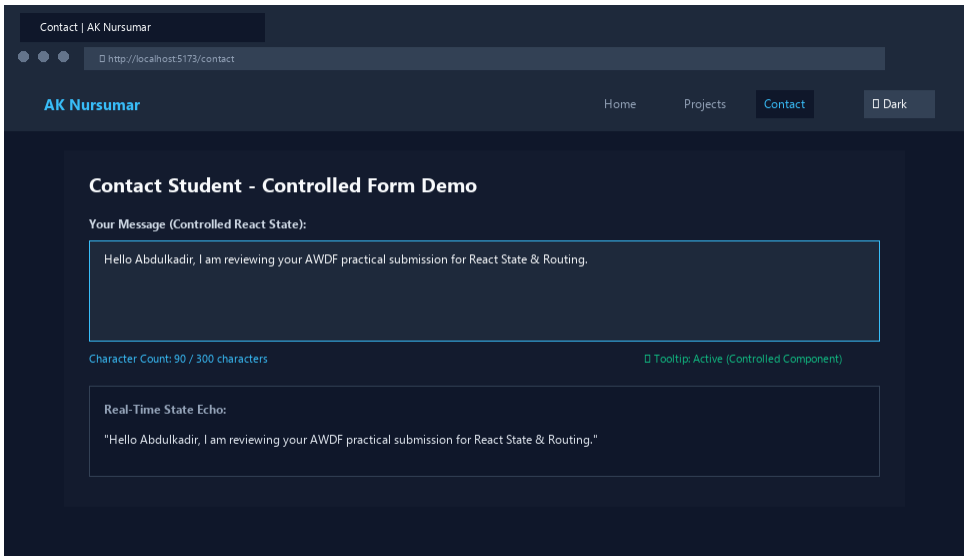
Explanation: *Wrapping App with BrowserRouter at the application entry point to synchronize UI routing with browser history.*

Figure 3: App.jsx – Route Definitions & Theme State



Explanation: *Declarative client-side route mapping inside App.jsx routing between pages without initiating full browser refreshes.*

Figure 4: Output – Controlled Contact Form with Live Character Count



Explanation: *Controlled textarea in the Contact route dynamically reflecting live keystrokes, character length (N / 300), and an interactive tooltip toggle.*

Key Questions and Answers

Q1. What is the role of each component in the overall UI structure?

In a routed Single Page Application (SPA), top-level components represent virtual pages (Home, Projects, Contact, NotFound). The NavBar acts as a persistent global navigation shell, while <Routes> dynamically unmounts the current page component and mounts the matching target component upon URL transitions without triggering a traditional HTTP page reload.

Q2. How does React re-render when props change?

When a component's props change due to state updates in an ancestor, React schedules a re-render. It recursively invokes the component's render function, produces a virtual element tree, diffs it against the preceding tree using its reconciliation heuristics, and commits only the updated DOM attributes or text nodes to the actual browser DOM.

Q3. Why is component reusability important in large-scale applications?

Component reusability enforces architectural consistency across complex applications. By isolating stateful behaviors (like controlled form fields, tooltips, or theme switchers) into reusable components, developers prevent duplicate code, ensure uniform styling and accessibility, and make unit testing straightforward.

Conclusion

Practical 2 successfully introduced client-side routing and reactive local state management into the portfolio application. React Router v6 was configured with four distinct routes, NavLink active state highlighting, a dark/light mode theme switcher, and a controlled contact form with live character counting.